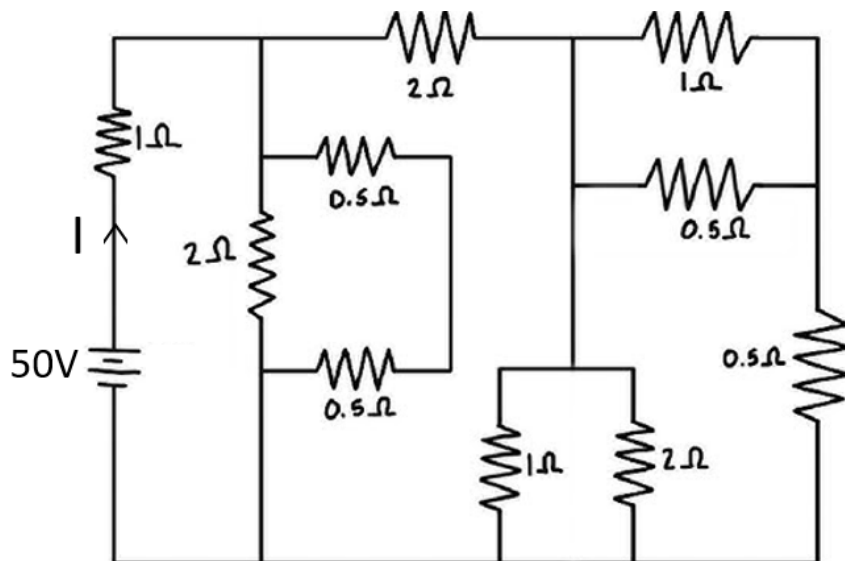
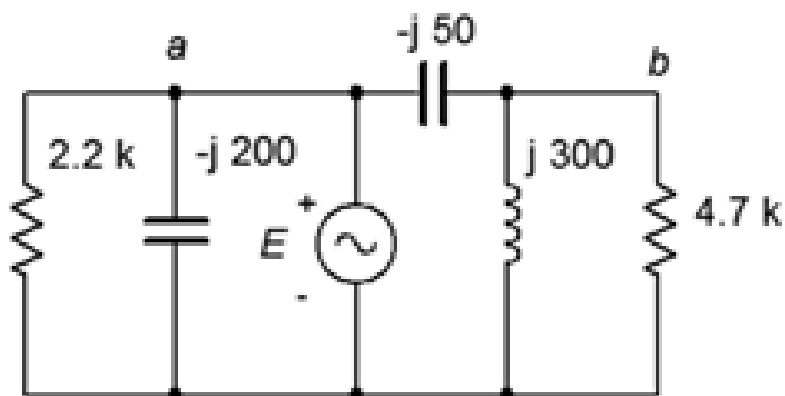


Exercises (AC-circuit)

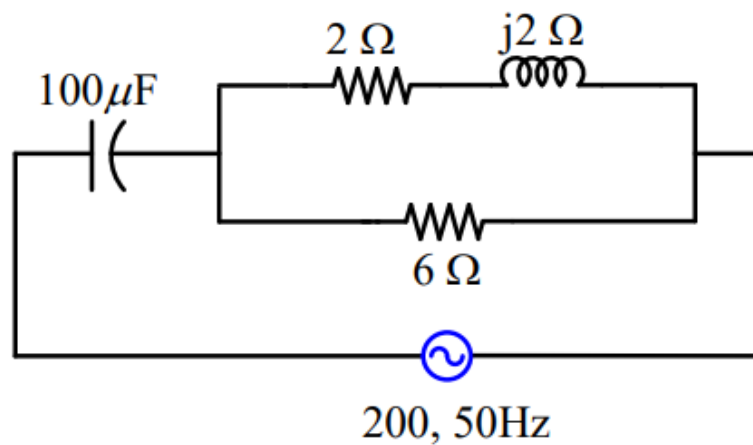
1. Find the equivalent resistance R_{eq} and current I ?



2. Determine the total impedance of the circuit? Draw a complex plane and find the angle?

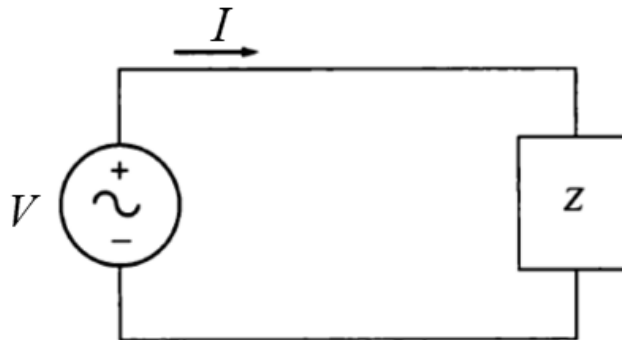


3. For the circuit shown in the following, find current in all branches?



4. Suppose we have an AC circuit with:

- Supply voltage: $V = 230\text{V (rms)}$
- Impedance of the load: $Z = 10 + j15\Omega$



- Find the magnitude of impedance?
- Find the current in rms?
- Find apparent power (S)?
- Find power factor?
- Find active power (P)?
- Find Reactive Power (Q)?